

Research Paper

Thermal prediction for Air-cooled data center using data Driven-based model

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ABSTRACT

The optimal cooling control of data centers (DCs) relies on the thermal model to simulate and accurately evaluate the temperature distribution of the computer room. Data-driven thermal modeling methods have been widely used in the thermal management of DCs. However, few existing works have comprehensively compared and analyzed the predictive performance and adaptability of various data-driven thermal models. In addition, most of the proposed thermal models are for temperature prediction in steady-state scenarios of DCs, ignoring the study of thermal changes in transient scenarios. Therefore, this work builds a CFD model of a typical data center as a verification platform to compare the temperature prediction performance of six machine learning (ML)-based thermal models (SVR, GPR, XGBoost, LightGBM, ANN, LSTM) in steady-state and transient-state scenarios. For steady-state scenarios, we also explore the impact of sample size, physical layout reconfiguration, and airflow pattern on performance to evaluate the adaptability of the thermal model. For transient scenarios, we design four cooling failure scenarios to verify the ability of the proposed thermal model to capture the dynamic thermal changes of the computer room. Numerous numerical simulation results show that the XGBoost-based and LightGBM-based thermal models outperform other ML models in both steady-state and transient-state scenarios, with prediction error RMSE is less than 1.0 °C, and the time overhead is less than 10⁻³ s. In addition, both thermal models are robust to physical layout reconfiguration and poor flow patterns.

1. Introduction

The cloud data center (CDC), as a physical platform for information system operation, has become an essential critical infrastructure for the operation of modern information society [1]. Meanwhile, massive data centers also bring enormous energy and carbon emissions [2]. In early 2020, Vafamehr et al. [3] published an article in Science claiming that the energy consumption and greenhouse gas (GHG) emissions of CDC account for 3% and 4% of the global total, respectively. The cooling system is an essential subsystem of the CDC, providing a safe thermal environment for IT equipment. However, it also accounts for more than 40% of the energy consumption [4]. Therefore, it is vital to conduct cooling management optimization research for energy efficiency and the safe operation of cloud data centers [5]. A key point of cooling optimization management is to quickly and accurately evaluate the temperature profile of the rack room to supply and allocate cold air while meeting the red-line temperature limit of the servers [6].

However, the complex geometries of data centers and the coupling of airflow and heat exchange within each rack pose significant challenges for developing thermal models for air-cooled DCs. Existing thermal modeling methods for DCs can be roughly divided into three categories, (1) Simplified physics-based models; (2) CFD-based models; and (3) Data-driven models (DDMs) [7]. A comparison of the computational overhead, model complexity, and accuracy of these three categories of models is shown in Fig. 1.

Thermodynamic-based models predict temperature distributions in extremely short computing times. They are often employed for parametric calculations and fast simulations of thermal profiles but with poor prediction accuracy. Thermal modeling based on CFD adopts computers to solve nonlinear partial differential equations of fluid flow to understand the state of fluid better qualitatively and quantitatively [8]. This method can provide complete and accurate thermal distribution prediction but requires many specific geometric parameters and a considerable numerical calculation. Therefore, this method is only suitable for the scheme evaluation in the design stage. It is difficult to be

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Nomenclature

Abbreviation

DCs	Data centers
CFD	Computational fluid dynamics
ML	Machine learning
SVR	Support vector regression
GPR	Gaussian process regression
XGBoost	eXtreme gradient boosting
LightGBM	Light gradient boosting machine
ANN	Artificial neural network
LSTM	Long short-term memory network
CDC	Cloud data center
GHG	Greenhouse gas
POD	Proper orthogonal decomposition
CRACs	Computer room air conditioners
DDMs	Data-driven models
GBDT	Gradient boosting decision tree
CART	Classification and regression trees
ACU	Air conditioner unit
PDU	Power distribution units
LHS	Latin hypercube sampling

MAE	Mean absolute error
MAPE	Mean absolute percentage error
MSE	Mean squared error
RMSE	Root mean square error
R^2	Coefficient of determination

Uppercase Letters

T_{Room}	The average temperature of the computer room (°C)
$T_{AC,SUP}$	ACU supply temperature (°C)
$T_{AC,SUP}$	ACU supply temperature (°C)
N_{AC}	ACU blower speed ratio(%)
T_{Rack_Inlet}	Rack inlet temperature (°C)
P_{Rack}	Rack power (kW)
X	The state space of the computer room

Subscript

t	Time step
i	Number of CRAC
j	Number of rack
h	The prediction horizon
w	Time window

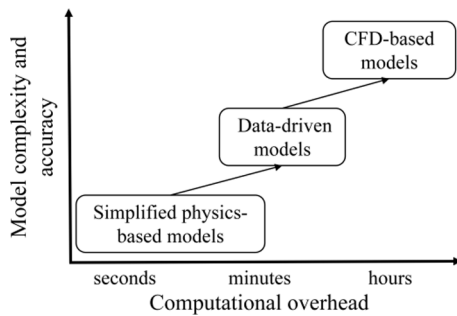


Fig. 1. Comparison of thermal models [5].

used for the rapid simulation and analysis of the thermal profile of DCs in the operation stage [9]. To achieve a receivable trade-off between the above two modeling approaches, data-driven models are proposed as a promising thermal modeling approach. The data-driven model captures the relationship between input characteristics and output target values by analyzing the collected data of the operating system without considering physical principles [7]. This approach can achieve a good trade-off between prediction accuracy and computational overhead, and has a good potential for real-time thermal management in large-scale DCs.

In recent years, significant progress has been made in data-driven thermal models due to the attention of all walks of life to the thermal management of DCs. Many related works have been published one after another. Ghosh et al. [10] used a data-driven statistical modeling framework with proper orthogonal decomposition (POD) to construct a thermal model of a data center. The proposed thermal model can accurately capture the characteristics of the two-dimensional state space of the rack thermal load and time to quickly and accurately predict the server room temperature. Experimental results show that the POD-based thermal model has high interpolation prediction accuracy but performs poorly for extrapolation prediction. In addition, artificial neural network (ANN) has gradually been widely used in the thermal modeling of DCs with its powerful nonlinear fitting ability [11]. Song et al. [12] developed an ANN-based thermal model based on CFD simulation data

to predict the inlet temperature of the server and the airflow of perforated tiles. The authors combine an ANN-based thermal model with a multi-objective genetic algorithm (GA) to design an optimization tool suitable for real-time thermal management in data centers. Similarly, Athavale et al. [13] conducted extensive numerical experiments to verify that the ANN-based thermal model is in good agreement with the CFD simulation model. Besides, this work evaluated the prediction accuracy and computational overhead of thermal models, which can guide the dynamic control of cooling systems. Moreover, Lloyd et al. [14] proposed a data-driven prediction model (DDPM) that can predict the server inlet temperature distribution in real-time, called the S-curve. Most notably, the work collected real temperature data from 52 different data centers and used clustering techniques to build a knowledge base that included temperature data and physical characteristics. Subsequently, this knowledge base is taken to train an ANN thermal model. The proposed ANN-based thermal model is not limited to the thermal prediction of a specific data center but can be generalized to the thermal management of different types of data centers. Zhang et al. [15] compared various real-time thermal prediction models based on machine learning, including ANN, Gaussian Process Regression (GPR) model, and Linear Regression (LR) model. These thermal models are employed to guide workload placement and migration on multi-node systems, reducing the cooling power of the entire computing system by 17% while maintaining performance. Surprisingly, the improved gradient boosting decision tree model (GBDT) with high computational efficiency and low computational overhead has gradually become a new choice for thermal modeling. Ilager et al. [16] collected sensor data from a private cloud, including resource utilization, host power, fan speed, etc., to train a temperature prediction model based on XGBoost. Guided by this thermal model, the authors developed a dynamic scheduling algorithm to minimize the peak temperature of the host. Note that the proposed method is universal and suitable for establishing data center temperature models with different physical environments and parameters.

Although various data-driven thermal models have been widely used in different scenarios and tasks of thermal management [17], some issues still have not been completely solved and need further exploration. First, the performance of data-driven thermal models depends on the quantity and quality of training samples. Still, few works have quantitatively analyzed the relationship between training sample size and

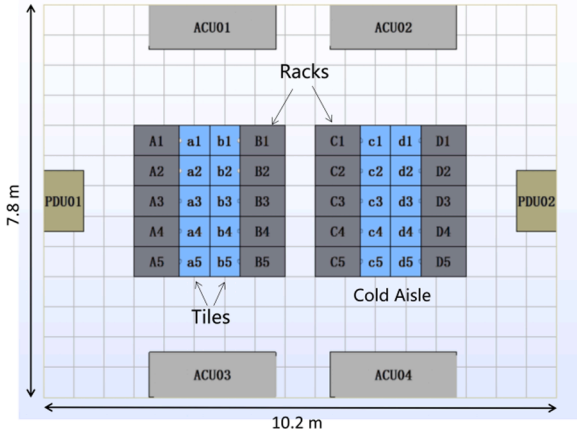


Fig. 2. CFD model.

performance. Secondly, the reconfiguration of the physical layout and the equipment state parameters will disturb the airflow pattern, leading to the degradation of the performance of the thermal model or even failure. Therefore, the adaptability of the thermal model to data center reconfiguration is worth further exploration. Finally, most of the existing proposed thermal models serve for temperature prediction in steady-state scenarios, while there are few studies on dynamic temperature changes in transient-state scenarios.

To fill this gap, we constructed a CFD model of a data center based on 6SigmaDC simulation software as an experimental platform for thermal model performance comparison. Furthermore, we conduct numerical experiments to explore the impact of sample size, data center reconfiguration, and physical layout on model performance. Besides, four common cooling failure scenarios are designed to evaluate the thermal model's ability to perceive temperature changes in the transient-state scenario. This work aims to comprehensively compare and evaluate the applicability and limitations of widely used and emerging thermal models in different scenarios and to provide guidance for selecting appropriate thermal models for DCs. In conclusion, the key contributions of this work are as follows:

1. We constructed the data center CFD model as a verification platform to comprehensively compare the temperature prediction performance of six machine learning-based thermal models (SVR, GPR, XGBoost [18], LightGBM [19], ANN, LSTM) in steady-state and transient-state scenarios.
2. We also explore how sample size, physical layout reconfiguration, and airflow patterns affect thermal model performance to evaluate model fitness.
3. For transient-state scenarios, we design four cooling failure scenarios to verify the ability of the thermal model to capture the thermal evolution.
4. Compared with other proposed thermal models, XGBoost and LightGBM have better performance, with RMSE less than $1.0\text{ }^{\circ}\text{C}$ and time overhead less than 10^{-3} s .

The remainder of this work is organized as follows. Section 2 introduces the hardware configuration of the data center, application scenarios, and data collection process in detail. Then, Section 3 describes the construction steps of the data-driven thermal model and the principles of the six machine learning models. Numerous numerical experiments and results will be introduced and analyzed in Section 4. The conclusion and potential future research directions are proposed at the end.

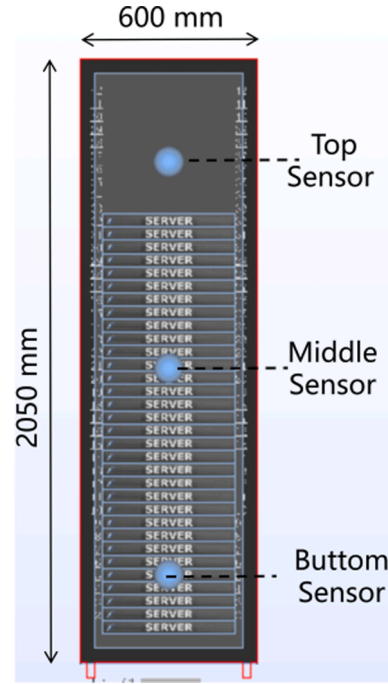


Fig. 3. Sensor deployment position.

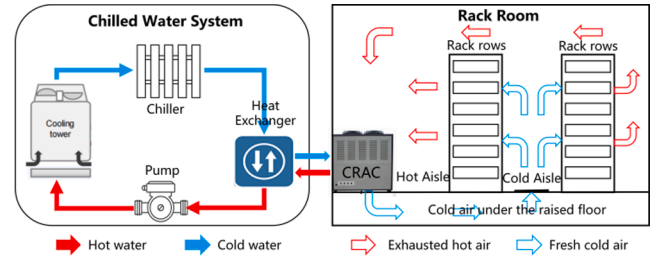


Fig. 4. Data center cooling system configuration.

2. Data center configuration and scenarios

The commercial simulation software 6SigmaDCX [20] was used to develop a room-level CFD model, as shown in Fig. 2. The detailed configuration of the proposed CFD model is presented in Section 2.1. Additionally, two simulation scenarios (steady-state and transient-state) are designed to verify the performance of the data-driven thermal model. A steady-state scenario means that the computer room eventually tends to a state of thermal equilibrium under specific state parameters. The transient scenario simulates the instantaneous state of the computer room at a certain moment during the dynamic change of the online state parameters [21]. The following sections describe the parameters considered and the dataset acquisition process for steady-state and transient-state scenarios.

2.1. Data center configuration

The top view of the computer room CFD model is shown in Fig. 2. The IT infrastructure is equipped with 20 racks arranged in 4 rows. The computer room adopts a raised floor for cooling, and an open cold aisle is arranged between the rack rows. Two power distribution units (PDUs) are used as power supply equipment to supply power to all racks. To collect inlet temperatures at different locations of the racks, three temperature sensors were deployed at the front air intakes of each rack, as shown in Fig. 3. In addition, the cooling system is equipped with four computer room air conditioners (CRACs) and a cooling water system

Table 1
Details of CFD Model.

Parameter	Value
Room size	10.2 m × 7.8 m × 4.5 m
Height of floor static pressure chamber	0.64 m
Number of rack	20
Rack Model	42U racks
Number of ACU	4
Number of PDU	2
Power density	8.52 kW/Rack
Air supply of air conditioner	9 m ³ /s
supply air temperature	20 °C
Tile Porosity	50%
Number of Grid Elements	1 × 10 ⁶
Grid type	Structured network
Turbulence Model	k-ε turbulence model

Table 2
Parameter Setting.

Independent variable	Range	Constraints	Dependent Variable
ACU Blower Speed (N_{AC})	40%-100%	Separated by 5% i.e.(40%, 45%,..., 100%)	60 static sensor temperatures
ACU Temperature Set Point ($T_{AC,SUP}$)	18 °C-27 °C	Integer	
Rack Power (P_{Rack})	6 kW-12 kW	Set the maximum power of each cabinet to 12 kW, the working load from 50% to 100%, 5% apart, i.e. (6 kW, 6.6 kW,..., 12 kW)	

(Fig. 4) which includes heat exchangers, cooling units, pumps, and cooling towers. More details about the CFD model are summarized in Table 1.

2.2. Scenario instructions

Considering the time dimension, the scenarios in the data center can be roughly divided into steady-state scenarios and transient-state scenarios [7]. The steady-state scenario considers that the computer room tends to be in a stable state under specific parameter settings (for example, specific workload configuration and air conditioning parameter settings). However, the transient-state scenario is more concerned with the dynamic change of parameter settings to predict the future system state. The temperature prediction model in the steady-state scenario is more suitable for solving workload scheduling optimization and host hotspot problems. Therefore, here, the prediction target is the server inlet temperature. It is more used for dynamic cooling control and thermal risk assessment [9]. Therefore, the concern is the temperature of the computer room. The specific definitions are as follows. Assuming that at time t , the average temperature of the computer room is $T_{Room}(t)$; the supply temperature, return air temperature and wind speed of the air conditioner are expressed as $T_{AC,SUP}(t)$, $T_{AC,RET}(t)$, $N_{AC}(t)$ respectively; the rack operating power is expressed as $P_{Rack}(t)=\{P_1(t), P_2(t), \dots, P_j(t)\}$. That is, the temperature distribution of the rack inlet $T_{Rack,Inlet}$ in the steady-state scenario at time t is defined as,

$$T_{Rack,Inlet}(t) = f(T_{AC,SUP}(t), N_{AC}(t), P_{Rack}(t)) \quad (1)$$

Besides, the temperature prediction problem under cooling failure scenario is defined as a multivariate time series prediction problem. Note that it is assumed that the heating power of the computer room remains constant during the cooling failure period. Therefore, the state space of the computer room at time t can be expressed as $X(t)=\{T_{AC,SUP}(t), T_{AC,RET}(t), N_{AC}(t), T_{Room}(t)\}$. That is, the temperature distribution of the computer room at time $t+h$ in the transient scenario can be defined as,

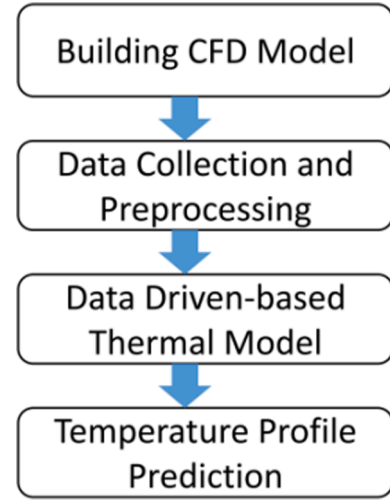


Fig. 5. The data-driven modeling approach.

$$T_{Room}(t+h) = f(X(t), X(t-1), \dots, X(t-w)) \quad (2)$$

where h represents the prediction horizon, that is, the temperature value after h time steps is predicted; w represents the historical sequence time window, that is, the historical sequence data of w time steps is used as the input feature of the model.

Steady-state scenario. Combined with the operation of the actual data center, the simulation uses adjustable independent variables including the fan speed N_{AC} of the CRAC, the cooling air supply temperature $T_{AC,SUP}$, and the operating power P_{Rack} of the rack. The dependent variable is the temperature set of 60 rack inlet sensors. The specific parameter settings are shown in Table 2. In order to make the independent variables of the sample as uniformly distributed as possible in the multi-dimensional parameter space and avoid the deviation of a certain dimension, the Latin hypercube sampling (LHS) method [22] was adopted to randomly generate 1100 sets of parameter combinations. Each parameter combination contains 28 parameters (wind speed N_{AC} , supply temperature $T_{AC,SUP}$ of 4 CRACs, and operating power P_{Rack} of 20 racks). Subsequently, the 6SigmaDCX [20] software was used for CFD numerical simulation calculation, and the temperature data records of all sensors $T_{Rack,Inlet}$ were derived. The dataset contains 1100 samples, each represented as a tuple $(N_{AC}, T_{AC,SUP}, P_{Rack}, T_{Rack,Inlet})$. Then, the dataset is divided into a training set and a test set according to the ratio of 10:1. The specific experimental setup parameters and steps are shown in Section 4.2.1.

Transient-state scenario. As a mission-critical facility, the data center will bring substantial economic losses and security risks in equipment failure and downtime [23]. Therefore, developing an accurate temperature prediction model for transient scenarios can guide thermal risk assessment and emergency measures in data centers [24]. As shown in Fig. 4, the cooling system components of DCs include the indoor CRAC and the outdoor chilled water system [25]. The chiller is responsible for cooling, the pump powers the circulating flow of coolant, and the cooling tower dissipates heat into the air in the form of gas evaporation [26]. Therefore, the main focus of this work is the cooling failure scenarios, including the failure of indoor CRAC and the failure of the cooling water system. CRAC failures are most likely due to unplanned power outages or downtime caused by the aging equipment. In addition, extreme high-temperature weather or local overheating of the outdoor unit will trigger the compressor or pump to trip and stop, failing the cooling water system [27]. Therefore, four common cooling failure sub-scenarios are designed for simulation verification. Here, we assumed that the total simulation time of all scenarios is 300 s, and the time step is one second, which can generate 300 sets of time-series data

with timestamps. Each group of samples is defined as a tuple $(t, T_{AC,SUP}, T_{AC,RET}(t), T_{Room}(t))$, the first 240 samples are the training set, and the remaining 60 samples are the test set. The specific experimental setup parameters and steps are shown in Section 4.2.2.

3. Data driven-based thermal prediction model

Data-driven models (DDMs) are dedicated to exploring the laws and relationships between system state characteristics and target variables from massive data without focusing too much on physical behavior [7]. The data-driven modeling approach is shown in Fig. 5. First, a CFD model is constructed according to the specific data center's physical layout and infrastructure configuration. Subsequently, numerical simulations were performed using CFD software to generate data and pre-processed. Next, the sample data is used to train the thermal model. Finally, the sensor data collected in real-time is input into the thermal model for temperature profile prediction.

Six representative data-driven models were selected for comparison and analysis, including nonlinear models - Support Vector Regression (SVR) models and Gaussian Process Regression (GPR) models, ensemble learning models - XGBoost [18] and LightGBM [19], Artificial Neural Network (ANN) [28] and Long Short-Term Memory Neural Network LSTM [29]. Note that LSTM is only used for time series prediction in transient scenarios. Furthermore, the thermal model needs to evaluate the inlet temperature of multiple servers, but the above partial regression model only supports a single output. Therefore, the wrapper MultiOutputRegressor in the Scikit-learn [30] library is adopted to create a separate model for each output of the problem, building a multi-output regression model. To minimize hyperparameters' impact on model performance, the grid search method provided by the toolkit GridSearchCV can be used to conduct an exhaustive search for the specified parameter values to obtain the optimal hyper-parameter set (see Appendix Table 6). Also, K -Fold cross-validation was employed to overcome model over-fitting, where K is 10. The six data-driven models employed in this work are briefly introduced below.

3.1. SVR

Support Vector Regression (SVR) [31] maps low-dimensional data to a high-dimensional feature space through a kernel function. It carries out linear regression in this space, encoding the nonlinear relationship between the independent variables of the input and the target value of the output. Given a set of sample data (x_n, y_n) , where x_n is the matrix of independent variables, y_n is the dependent variable, and N is the total number of samples. For general regression problems, SVR wants to learn a function $g(x_n)$ that is as close to y_n as possible, and $g(x)$ can be expressed as:

$$g(x) = \sum_{n=1}^N w_n \varphi_n(x) + b \quad (3)$$

where $\varphi_n(x)$ is a mapping function of data from low dimension to high dimension, and w_n and b are the parameters to be determined. The optimization objective of the SVR is to minimize the loss function:

$$P(g(x)) = \frac{C}{N} \sum_{n=1}^N E_e(y_n, g(x_n)) + \frac{\|w\|^2}{2} \quad (4)$$

where $E_e(y_n, g(x_n))$ is a differentiable loss function representing the deviation between the evaluated value $g(x_n)$ and the observed value y_n ; $\frac{\|w\|^2}{2}$ is the planarity term; C is the regularization constant, which is used to prevent the model from overfitting; For a more detailed description of SVR, please refer to [31].

3.2. Gpr

GPR is a widely used machine learning model based on Gaussian Process (GP) prior distribution. Due to the convenient nature of GP and kernel functions, GPR is often used to solve nonlinear regression problems. GP is one of the stochastic processes, which is a collection of normally distributed random variables within an index set. If $\{f(x), x \in R^d\}$ is a GP, and n observations $(x_1, x_2, \dots, x_n)$ are given, then the random variables $(f(x_1), f(x_2), \dots, f(x_n))$ also satisfy the Gaussian distribution. Given a set of sample data (x_n, y_n) , a standard linear regression model with Gaussian noise can be expressed as,

$$y = x^T \beta + \varepsilon \quad (6)$$

where x is the input matrix and y is the observed value. $\varepsilon \sim N(0, \sigma^2)$, the error variance σ^2 and the coefficient β can be estimated from the sample data provided. A more detailed description of GPR can be found in Ref. [32].

3.3. Xgboost

XGBoost¹ (eXtreme Gradient Boosting) is a modified gradient boosting decision tree (GBDT) algorithm that is widely used in classification and regression problems [18]. It uses Newton's method in solving the extremum of the loss function, which expands the loss function, Taylor, to the second order. XGBoost is widely used in data mining competitions and large-scale production systems in the industry. XGBoost is an additive model based on the idea of boosting, which combines K classification and regression trees (CART). Given a dataset containing n instances, each instance contains m features. Take the feature matrix $x_i = \{x_{i1}, x_{i2}, \dots, x_{im}\}$ of instance i as the input of the model, and output the predicted value $\hat{y}_i$, which is expressed as,

$$\hat{y}_i = \Phi(x_i) = \sum_{k=1}^K f_k(x_i), f_k \in F \quad (7)$$

where F represents the space of all regressors and f_k represents an independent tree. The goal of XGBoost is to explore the set of functions employed by the model to minimize the objective loss function:

$$L(\Phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \quad (8)$$

$$\text{where } \Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$$

where l is a differentiable loss function representing the deviation between the predicted value $\hat{y}_i$ and the true value y_i ; the regular term Ω is to prevent over-fitting of the model; T is the number of leaves in the tree; w is the weight of each leaf node. A more detailed description of XGBoost can be found in Ref. [18].

3.4. LightGBM

Light Gradient Boosting Machine (LightGBM²) is another extended version of the GBDT model. Compared with Xgboost, it performs better in training speed and memory usage [19]. Numerous practices have proved that LightGBM can solve the time-consuming problem of GBDT processing large sample of high-dimensional data, making GBDT more suitable for real industrial systems. LightGBM performs four optimizations on the traditional GBDT ensemble algorithm. The first is to use the histogram algorithm to determine the optimal segmentation point. Secondly, in order to figure out the problem of large data volume, the gradient-based one-side sampling (GOSS) method only uses data

¹ <https://github.com/dmlc/xgboost>.

² <https://github.com/Microsoft/LightGBM>.

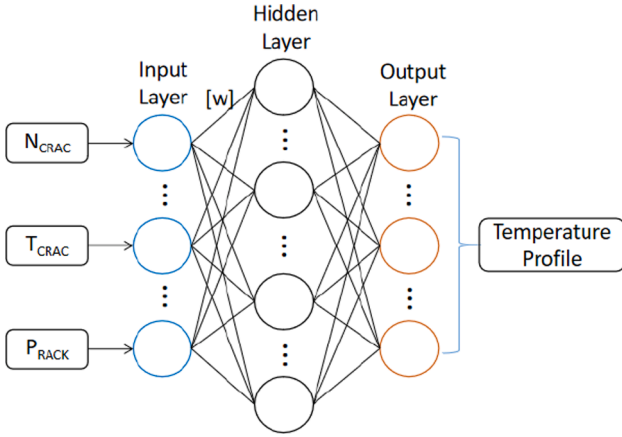


Fig. 6. Common structure of ANN.

instances with noticeable gradient changes to estimate the information gain, which effectively saves the computational time and space overhead. Moreover, the use of Exclusive Feature Bundling (EFB) bundles multiple contradictory features into one feature to achieve dimensionality reduction. Finally, a leaf-wise leaf growth strategy with depth restriction is adopted. A more detailed description of LightGBM can be found in Ref. [19].

3.5. Ann

Artificial neural network (ANN) is a widely adopted data-driven modeling method. The powerful nonlinear input–output fitting capabilities of ANNs are particularly appropriate for the thermal modeling of DCs in complex environments [28]. The key challenge facing ANN is the trade-off between model complexity and performance. Over-fitting can occur when the network parameters are much larger than the available training samples. Otherwise, poor-fitting may occur.

A common ANN model includes a three-layer structure (Fig. 6), where the input layer receives feature vectors or raw pixel data; the output layer outputs the value vector after multi-layer calculation; the hidden layer is the connection between the input layer and the output layer, and its function is to convert the input features into more abstract features for better linear division. The mathematical definition of the network structure is expressed as,

$$y_j = f\left(\sum_{i=1}^n w_{ij}x_i + b_j\right) \quad (9)$$

where x_i is the feature vector. Specifically, the input of the thermal model is the air-conditioning wind speed N_{AC} , supply temperature $T_{AC,SUP}$ and rack power P_{RACK} in the computer room. The ANN model updates the parameter w through the back-propagation method, and its loss function is determined by the mean square error (MSE), which is expressed as,

$$MSE = \frac{1}{N} \sum_{n=1}^N (\hat{T}_n - T_n)^2 \quad (10)$$

where N is the number of given samples, $\hat{T}_n$ and T_n denote the predicted temperature and the actual temperature, respectively. In addition, adding a dropout layer can effectively strengthen the generalization ability of the model. Also, the adaptive learning rate method can speed up the training convergence [33]. The hyper-parameter settings of the model are shown in Appendix Table 6.

3.6. Lstm

Long Short Term Memory Network (LSTM) is a modified recurrent

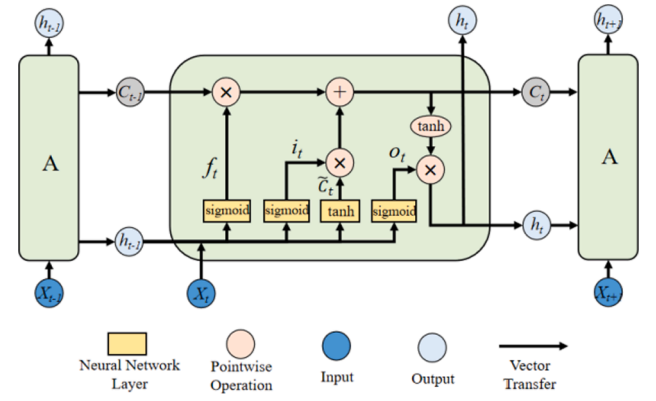


Fig. 7. The basic structure of LSTM.

neural network (RNN) that can tackle problems with long-distance dependencies [29]. The basic structure of LSTM, as shown in Fig. 7, consists of multiple cells connected to a state chain. The current cell state C_t is jointly affected by the current input X_t , the previous cell state C_{t-1} and the output h_{t-1} , forming a certain memory function for the time series. Furthermore, each cell consists of three control gates, where f_t is the forget gate, i_t is the input gate, and o_t is the output gate. Gates are a way of selectively letting information through, and they consist of a sigmoid network layer and pointwise multiplication operations. LSTM controls whether the previous network memory state acts on the current network through three gates on the cell. The specific calculation process of LSTM is as follows [34].

The first step is to decide what information will be retained from the previous cell state to the current cell state through the sigmoid layer of the forget gate, which is expressed as follows,

$$f_t = \text{sigmoid}(W_f \cdot [h_{t-1}, X_t] + b_f). \quad (11)$$

The second step is to go through an input gate to decide which values will be updated. Here, a new cell state candidate value vector needs to be generated by the tanh layer, which is expressed as,

$$\tilde{C}_t = \tanh(W_c \cdot [h_{t-1}, X_t] + b_c) \quad (12)$$

Then, through the sigmoid layer of the input gate to decide which values to update,

$$i_t = \text{sigmoid}(W_i \cdot [h_{t-1}, X_t] + b_i) \quad (13)$$

Finally, the cell state is updated from C_{t-1} to C_t , which is expressed as,

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (14)$$

The final step is to decide the output of the model. First, an initial output is obtained through the sigmoid layer of the output gate, then the value is scaled to $[-1, 1]$ using tanh, and finally multiplied by the output o_t to obtain the output h_t of the model, expressed as follows:

$$o_t = \text{sigmoid}(W_o \cdot [h_{t-1}, X_t] + b_o) \quad (15)$$

$$h_t = o_t * \tanh(C_t) \quad (16)$$

where W_f , W_i , W_o , b_f , b_i , b_o represent the weight and bias of each network layer, respectively.

4. Simulation setup and result analysis

4.1. CFD simulation setup

The simulation experiment includes two objectives. The first is to compare and analyze the performance of thermal models in steady-state and transient scenarios. Secondly, the influence of physical layout and airflow circulation on model performance is explored. Therefore, the simulation experiment is divided into two parts: the steady-state

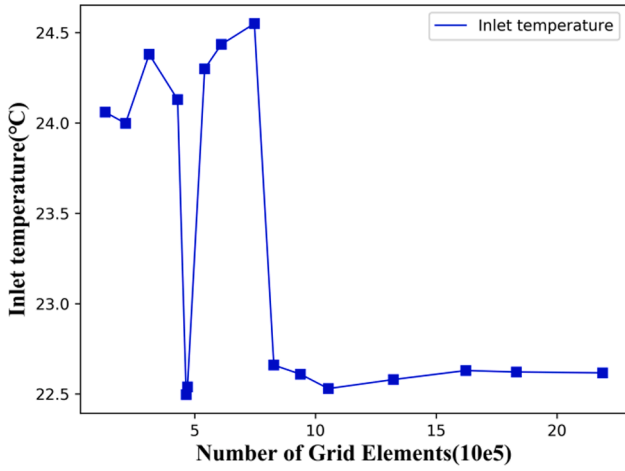


Fig. 8. Intel temperature changes with the number of grid elements.

scenario and the transient scenario. The specific experimental steps and results under different scenarios are described in Section 4.2.

The number of grids in the CFD simulation is critical to the calculation results. Therefore, to minimize the computational error of the simulation, we performed grid independence verification to find the appropriate number of grids for computation. The critical parameter server inlet temperature is selected as the judgment parameter for grid independence verification. As shown in Fig. 8, when the number of grids reaches 10^6 , increasing the number of grids will no longer affect the change of this parameter. Thus, it can be judged that the number of 10^6 grids has met the calculation requirements to a certain extent. In addition, the CFD simulations set the maximum number of iterations to 1000 and assume that the solution converged when the residuals of the variables between successive iterations were less than 10 ppm.

Referring to the previous work [15,16], the simulation experiments use error metrics commonly used to evaluate prediction models, including Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), Mean Squared Error (MSE), Root Mean Square Error (RMSE) and coefficient of determination (R^2). R^2 measures the fit between the estimate and the actual value. The value range of R^2 is [0, 1], and the closer the value is to 1, the higher the fitting degree is. The calculation equation of these metrics is respectively defined as,

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{T}_i - T_i| \quad (17)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{T}_i - T_i}{T_i} \right| \times 100\% \quad (18)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{T}_i - T_i)^2} \quad (19)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (\hat{T}_i - T_i)^2 \quad (20)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (T_i - \hat{T})^2}{\sum_{i=1}^n (T_i - \bar{T})^2} \quad (21)$$

where $\hat{T}_i$ is the estimate value, T_i is the actual value, and $\bar{T}$ is the average value. The commercial CFD simulation software 6SigmaDCX was used to model the data center and collect data. Moreover, the training and testing experiments of all models are carried out under the programming environment of Python-3.6 and Pytorch-1.0.0. The hardware device is a desktop computer with Windows 10 operating system, Intel Core i7-6700HQ processor, 12G memory, and an NVIDIA GeForce GTX 940MX

Table 3

Error metrics.

Model	MAE (°C)	MAPE (%)	MSE (°C)	RMSE (°C)	R^2
SVR	0.883	3.377	1.600	1.265	0.770
GPR	0.817	3.150	1.290	1.136	0.807
XGBoost	0.712	2.741	1.022	1.011	0.859
LightGBM	0.692	2.652	1.006	1.003	0.862
ANN	0.806	3.120	1.235	1.111	0.815

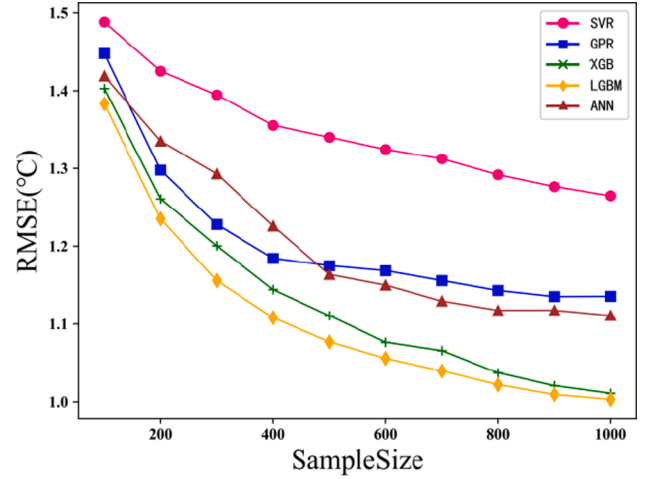


Fig. 9. Relationship between sample size and prediction error.

graphics card.

4.2. Result analysis

4.2.1. Steady-state scenario

The 1100 sample data collected by the CFD simulation were divided into 1000 training sets and 100 testing sets to train five data-driven thermal models. The above five error metrics are adopted to evaluate the model's overall performance. From Table 3, it can be clearly observed that the two thermal models based on XGBoost and LightGBM outperform the other models with RMSE close to 1 °C and MAPE less than 3.0%.

The data-driven model requires a large amount of data to learn the feature space. However, it is difficult to obtain sufficient data in a realistic operating environment due to the extended data collection cycle in the computer room [9]. Therefore, exploring the influence of sample size on the prediction accuracy of data-driven models can help guide the selection of appropriate modeling methods and give estimates of expected errors. Fig. 9 shows that the prediction error of each model decreases monotonically with increasing sample size until convergence. In addition, the prediction errors of XGBoost and LightGBM are smaller than other models under various sample sizes, and the RMSE value is close to 1.0°C when the sample size reaches 1000. Note that the ANN model outperforms the GPR model after the sample size exceeds 500. From this, it can be inferred that the performance of ANN models with extensive network parameters is more dependent on the amount of training data.

Since the IT load in the data center changes dynamically, managers often adopt server consolidation technology to achieve energy-saving optimization by shutting down some IT equipment and choosing to shut down some indoor CRACs [35,36,37]. These two energy-saving measures will significantly change the computer room's airflow circulation and heat distribution. Therefore, the thermal model must maintain prediction accuracy in the reconfiguration scenario. Two types of data center reconfiguration modes are designed here: setting the power

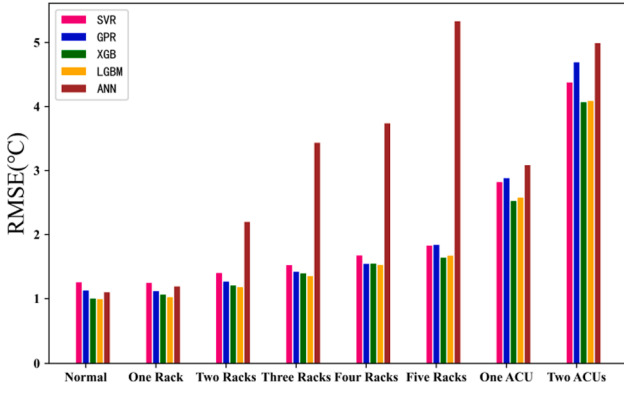


Fig. 10. Relationship between prediction error and configuration mode.

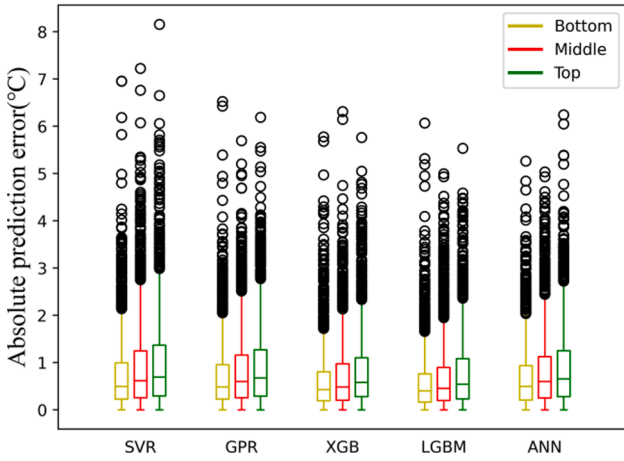


Fig. 11. Relationship between sensor location and prediction error.

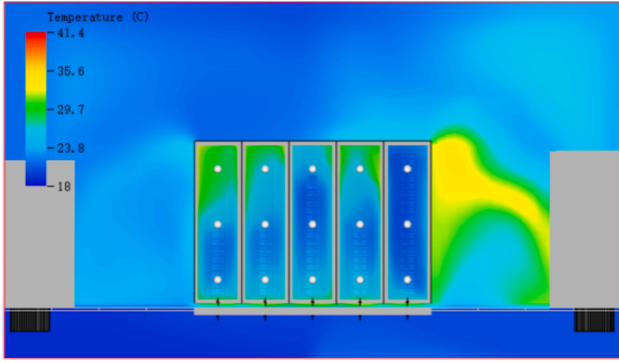


Fig. 12. Temperature profile of racks.

of 1 to 5 racks in the machine room to 0 kW and shutting down 1 to 2 indoor CRACs, respectively. For the seven sub-modes, 50 additional CFD simulation samples were collected for each sub-mode as test data. Fig. 10 shows the change in model prediction error under seven data center reconfiguration modes. Obviously, with the increase in the number of closed racks, the prediction error of most models increases slightly. However, the ANN-based thermal model has poor adaptability, and the RMSE increases significantly. Moreover, shutting down the CRAC has an even more significant impact on the prediction error. From this, it can be concluded that both rack power fluctuations and shutting down air conditioners can cause significant changes in the airflow and temperature of the rack room, which directly affect the model's prediction accuracy. Besides, the degree of influence of air conditioner state

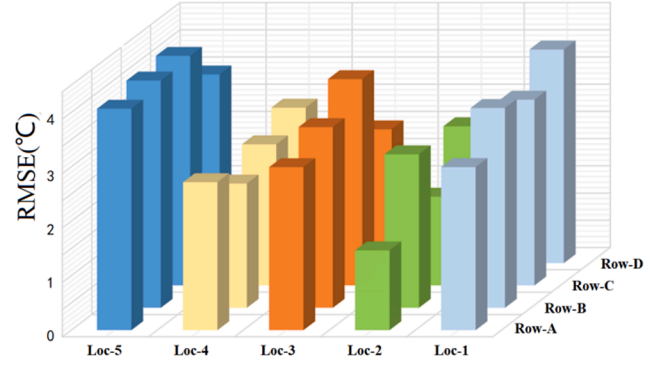


Fig. 13. Relationship between rack location and prediction error.

Table 4

Parameters of cooling failure scenario.

Scenarios	Sub-scenarios	Parameter Setting
CRAC Failure	Scenario-1: One CRAC Failure	CRAC ID:ACU01
		CRAC Restart Time:120 s
		CRAC Slow Down Time:60 s
		CRAC Starting Time:30 s
Chilled Water System Failure	Scenario-2: All CRACs Failure	CRAC ID:ACU01, ACU02, ACU03, ACU04
		CRAC Restart Time:120 s
		CRAC Slow Down Time:60 s
		CRAC Starting Time:30 s
Chilled Water System Failure	Scenario-3: Chiller Failure	None
	Scenario-4: Pump Failure	Pump Restart Time:120 s

changes is much greater than that of rack power fluctuations.

Thermal recirculation is a common airflow pattern in an air-cooled data center that causes localized airflow disturbances and temperature distribution gradients [38]. To explore the relationship between sensor location and prediction error, we deployed three temperature sensors at different rack heights (see Fig. 3). The boxplot (Fig. 11) shows that as the sensor deployment height increases, the absolute prediction error of the model is larger, and the distribution is more spread out. In addition, from the temperature profile of a row of racks (Fig. 12), it can be seen that the hot air recirculation at the top of the rack is more severe than that in the middle and bottom. Therefore, it can be inferred that thermal recirculation makes it difficult for the data-driven model to accurately capture the temperature variation characteristics, thus affecting the prediction accuracy.

To explore the relationship between rack location and prediction error, the temperature prediction errors of racks in different locations are analyzed and counted. Fig. 13 shows the prediction error of the LightGBM model. The prediction error RMSE at both ends of the rack row (Loc-1, Loc-5) and the middle location (Loc-3) is significantly larger than Loc-2 and Loc-4. There are two reasons for this phenomenon. First, the cold aisle between the two rows of racks is open, so the hot air returning from the side effects the two ends. In addition, multiple cold airflows from different air conditioners converge in the middle location (Loc-3), leading to the complexity and instability of the airflow, thus losing the model's prediction accuracy.

4.2.2. Transient-state scenario: Cooling failure

Once cooling failure occurs in the data center, the temperature of the rack room will rise sharply, leading to IT equipment downtime [27] or thermal risk. Therefore, it is essential to evaluate the temperature change of the room when the air-conditioning system fails, which has guiding significance for the system design, operation and maintenance, and emergency plan of the data center. Here, four sub-scenarios are

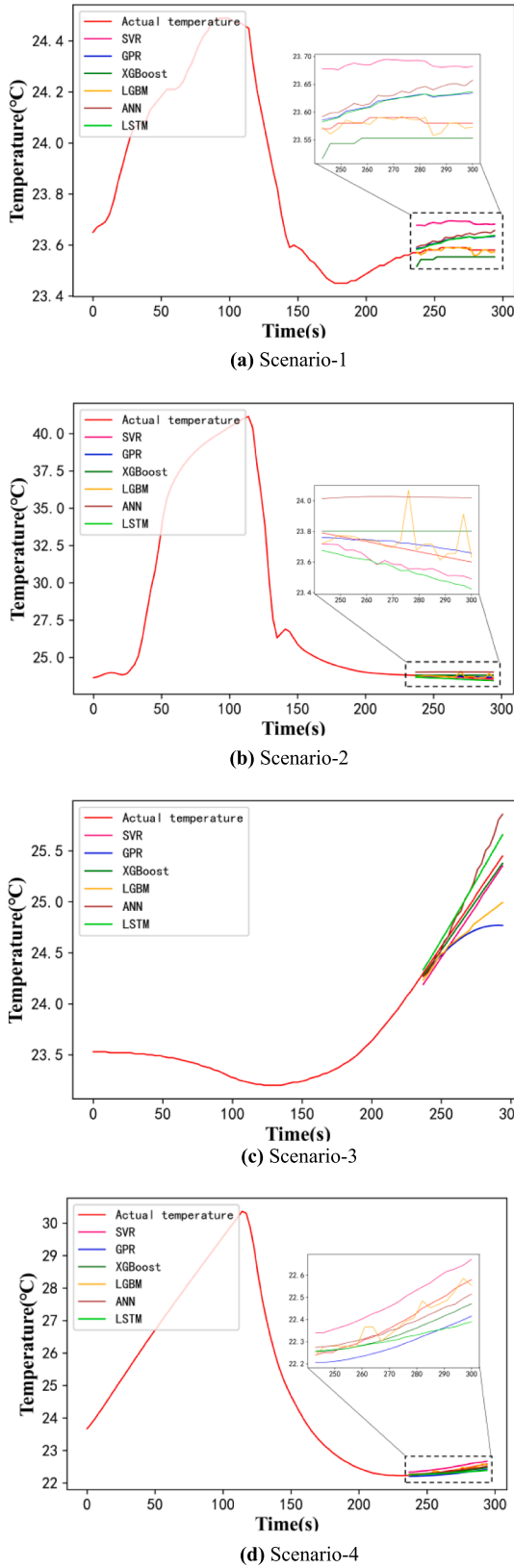


Fig. 14. Temperature prediction under four cooling failure scenarios.

designed to verify the model's performance under cooling failure scenarios. The specific simulation parameter settings are shown in Table 4.

Fig. 14(a-d) shows the temperature prediction errors of various models under the four cooling failure sub-scenarios. It can be observed

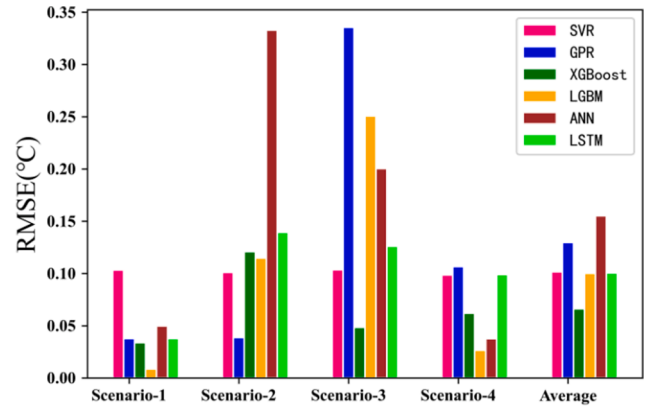


Fig. 15. RMSE of the prediction model under four sub-scenarios.

Table 5

Computational overhead.

Model	Training time (s)		Calculation time (s)		Memory usage (MiB)	
	SS	TS	SS	TS	SS	TS
SVR	4.5	0.004	5.3×10^{-3}	2.9×10^{-5}	18.8	0.2
GPR	472.1	0.320	4.3×10^{-3}	4.6×10^{-5}	319	2.5
XGB	97.2	0.466	9.0×10^{-3}	4.5×10^{-4}	76.1	0.9
LGBM	5.3	0.268	1.3×10^{-3}	6.9×10^{-5}	8.2	4.4
ANN	47.0	12.473	3.5×10^{-3}	6.1×10^{-4}	4.2	2.2
LSTM	–	18.158	–	1.9×10^{-3}	–	22.8

SS: Steady-state scenario; TS: Transient-state scenario.

that each data-driven model can generally capture the changing trend of computer room temperature. Note that in scenarios with severely insufficient cooling supply (Scenario-2, Scenario-3), the thermal model is disturbed by complex environmental factors, increasing prediction errors. On the whole (see Fig. 15), XGBoost, LightGBM, and LSTM models perform well in time series prediction scenarios, with an average RMSE value below 0.15°C. Conversely, the prediction errors of the SVR and GPR models are more volatile.

Computational overhead is also one of the essential metrics to measure whether the model has practical application value. The thermal model deployed on the real-time operating system not only needs high prediction accuracy but also the calculation time needs to reach the standard of real-time calculation. Therefore, three attributes of the model's training duration, prediction time, and memory overhead are taken to measure the computational overhead. The experimental results (Table 5) show that all thermal models can be calculated in an extremely short time (less than 10^{-3} s), suitable for real-time prediction. Also, except for the GPR model, other thermal models occupy fewer system memory resources. In general, the LightGBM thermal model has the superiorities of high prediction accuracy, short training time, and less memory usage, so it has a good application prospect.

5. Conclusion

To select suitable thermal models for data centers, this work systematically compares and analyzes six data-driven thermal models (SVR, GPR, XGBoost, LightGBM, ANN, LSTM) in steady-state and transient-state scenarios. The experimental results show that the ensemble learning algorithm (XGBoost, LightGBM) is superior to other comparison algorithms in prediction accuracy, computational overhead, and adaptability. Neural network-based models depend more on the quantity and quality of training samples. In addition, numerous simulation experiments are adopted to reveal the influence of the layout of IT equipment and the thermal recirculation airflow on the model accuracy,

Table 6
Computational overhead.

Model	Hyper-parameter
SVR	C: 100; Gamma: 0.01
XGBoost	Learning rate: 0.03; Max depth: 5 Min child weight: 5; Gamma: 0.1 Colsample bytree: 0.4
ANN	ANN model Configuration: Steady-state: 28–100-60, Transient-state: 18–50-1; Activation Function: ReLU; Optimizer: Adam; Dropout rate: 0.5 Batch size: 12; Learning rate: 0.01
GPR	Alpha: 0.05; N_restarts_optimizer: 2
LightGBM	Learning rate: 0.1; N_estimators: 80 Num_leaves: 10; Boosting_type: gbd
LSTM	LSTM model configuration: 18–80-1; Optimizer: Adam; Batch size: 12 Learning rate: 0.001; Lambda_l2: 0.0001

which guides optimizing the airflow management of the rack room.

Therefore, based on this work, the follow-up research direction can explore how to use auxiliary equipment to optimize the airflow organization of the computer room, such as installing fans on active perforated tiles or inter-row fans. Furthermore, grey-box modeling, as a novel modeling approach that fuses the laws of physics and data-driven methodologies, is a promising direction for building thermal models of data centers. Finally, we will conduct research on thermal-aware IT load scheduling and dynamic cooling control based on the established thermal model in the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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Appendix

See Table 6.

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